UNIT 2 · 9 hours

DC Motor: Principle of Operation and Back EMF



Transformers and Electrical Machines

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- Describe the construction of a DC machine
- Explain the principle of the DC motor (Fleming's left-hand rule)
- Derive the voltage equation and explain back EMF
- Explain the significance of back EMF, including starting

Syllabus context

DC Motor: principle of operation, back EMF and its significance, torque equation, characteristics and speed control (armature and field) of DC shunt motor.

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Construction of a DC machine

Stator

Yoke (cast steel frame), laminated pole cores with field coils, pole shoes to spread flux.

Rotor (armature)

Laminated silicon-steel core with slots; armature winding (lap or wave).

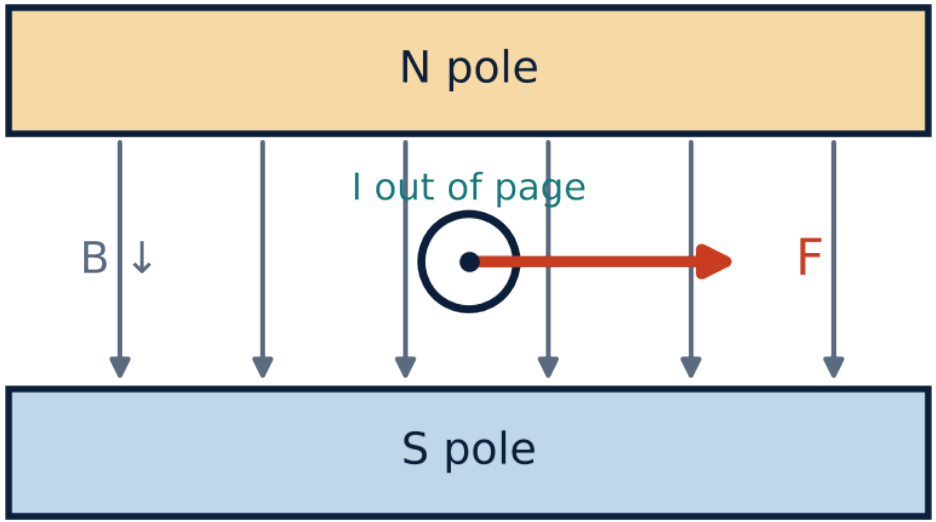
Commutator + brushes

Copper segments insulated by mica; carbon brushes carry current to the rotating armature.

Same construction serves as generator or motor. Lap winding: A = P; wave winding: A = 2.

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Principle of operation



- A current-carrying conductor in a magnetic field experiences a force.
- Direction: Fleming's left-hand rule (field, current, force).
- F = B I l.
- Forces on the two sides of a coil give a turning moment; the commutator keeps the torque in one direction.

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Back EMF and voltage equation

$$E_b = Z N \varphi P / (60 A)$$

back EMF (V)

$$V = E_b + I_a R_a$$

voltage equation

$$I_a = (V - E_b) / R_a$$

armature current

- Rotating conductors cut flux and induce an EMF opposing the applied voltage (Lenz's law): Eb.
- Eb ∝ φ N and is always less than V in motoring.

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Power relations

$$V I_a = E_b I_a + I_a^2 R_a$$

multiply by Ia

$$P_{in} = V I_a$$

electrical input to armature

$$P_m = E_b I_a$$

mechanical power developed

- $I_a^2 R_a$ is the armature copper loss.
- Efficiency of the armature conversion = Eb / V.

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Significance of back EMF

Self-regulation

If load rises, speed falls, Eb falls, Ia rises and torque increases: motor adjusts itself.

Starting

At standstill Eb = 0, so Ia = V/Ra is very large (≈ 20× full-load). A starter resistance limits it.

Energy conversion

Eb Ia is the electrical equivalent of mechanical power: Eb makes the conversion possible.

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Numerical: back EMF of a shunt motor

Problem A 220 V DC shunt motor takes 10 kW. Rsh = 230 Ω, Ra = 0.28 Ω. Find line, field and armature currents and back EMF.

- $I = 10\,000/220 = 45.45$ A
- $I_{sh} = 220/230 = 0.956$ A
- $I_a = I - I_{sh} = 44.49$ A
- $E_b = V - I_a R_a = 220 - 44.49 \times 0.28$

Answer: $I_a = 44.49$ A, $E_b = 207.54$ V

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Key takeaways

- Motor action: current-carrying conductor in a field experiences a force (Fleming's left hand)
- $V = E_b + I_a R_a$; $E_b = ZN\varphi P/60A$
- $E_b I_a$ = mechanical power developed
- Back EMF limits armature current; a starter is needed

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Practice questions

Q1. Why is a starter necessary for a DC motor?

Q2. A 240 V motor has Ra = 0.5 Ω and Ia = 20 A. Find Eb and mechanical power.

Q3. State Fleming's left-hand rule.

Q4. How does back EMF make a DC motor self-regulating?